

# Justin Payan Curriculum Vitae

Carnegie Mellon University  
Machine Learning Department  
Pittsburgh, PA

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## Employment

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10/2024 – **Carnegie Mellon University**  
*present* Postdoctoral Research Associate with Nihar Shah  
Pittsburgh, PA  
Current projects include greaptcha.com and auditing AI science workflows for vulnerability to backdoor attacks.

## Education

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09/2018 – **University of Massachusetts Amherst**  
02/2025 Ph.D. in Computer Science  
Advisor: Yair Zick  
09/2013 – **University of Georgia**  
05/2018 M.S. in Artificial Intelligence, B.S. in Mathematics, A.B. in Cognitive Science  
Advisor: Frederick Maier

## Past Employment

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05/2023 – **Adobe Research**, Document Intelligence Lab  
08/2023 Research Intern with Chris Tensmeyer  
06/2022 – **Microsoft**, Calc Intelligence  
09/2022 Data Science Intern with Kartik Sridhar  
06/2021 – **Amazon Alexa**, Trustworthy Alexa  
09/2021 Research Intern with Yuval Merhav  
05/2020 – **Amazon Alexa**, Trustworthy Alexa  
08/2020 Research Intern with Yuval Merhav  
06/2017 – **Vertica**, Machine Learning  
06/2018 Software Engineer with Vincent Xu

## Awards, Grants, and Scholarships

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2026 – 2027 **Alfred P. Sloan Foundation**  
Co-PI, “Reliable Autonomous AI Scientists.”  
With Nihar B. Shah and Atoosa Kasirzadeh.  
Spring 2026 **Google Cloud Computing Credits**  
Equivalent of \$30,000.  
Spring 2023 **UMass CICS Proposal Writing Grant**  
Amount: \$12,733.80.  
2013 – 2017 **University of Georgia Foundation Fellowship**  
\$128,260 in tuition and stipend for housing, research, and travel.

## Peer-Reviewed Publications

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Google Scholar: [https://scholar.google.com/citations?user=gc9w\\_eYAAAAJ&hl=en](https://scholar.google.com/citations?user=gc9w_eYAAAAJ&hl=en)

\* denotes equal contribution; ( $\alpha - \beta$ ) denotes alphabetical author order.

## Journal Publications

- [J1] Hadi Hosseini, Andrew McGregor, **Justin Payan**, Rik Sengupta, Rohit Vaish, and Vignesh Viswanathan ( $\alpha - \beta$ ). Graphical House Allocation with Identical Valuations. *JAAMAS 2024*. [link]

## Conference Publications

- [C8] **Justin Payan**, Noemi Barbagli, Ignacio Toledo, Rodrigo A. Carrasco, Sergio Martín, and Nihar B. Shah. Uncertainty-aware Ground-based Telescope Observation Scheduling at ALMA. *SPIE Astronomical Telescopes and Instrumentation 2026*. [link]
- [C7] Elita Lobo\*, **Justin Payan**\*, Cyrus Cousins, and Yair Zick. Fair and Welfare-Efficient Constrained Multi-matchings under Uncertainty. *NeurIPS 2024*. [link]
- [C6] **Justin Payan**, Swaroop Mishra, Mukul Singh, Carina Negreanu, Christian Poelitz, Chitta Baral, Subhro Roy, Rasika Chakravarthy, Benjamin Van Durme, and Elnaz Nouri. InstructExcel: A Benchmark for Natural Language Instruction in Excel. *EMNLP 2023 (Findings)*. [link]
- [C5] Cyrus Cousins, **Justin Payan**, and Yair Zick ( $\alpha - \beta$ ). Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities. *SAGT 2023*. [link]
- [C4] **Justin Payan**, Rik Sengupta, and Vignesh Viswanathan ( $\alpha - \beta$ ). Relaxations of Envy-Freeness over Graphs. *AAMAS 2023 (Extended Abstract)*. [link]
- [C3] **Justin Payan** and Yair Zick. I Will Have Order! Optimizing Orders for Fair Reviewer Assignment. *IJCAI 2022*. [link]
- [C2] **Justin Payan**, Yuval Merhav, He Xie, Satyapriya Krishna, Anil Ramakrishna, Mukund Sridhar, and Rahul Gupta. Towards Realistic Single-Task Continuous Learning Research for NER. *EMNLP 2021 (Findings)*. [link]
- [C1] Ananya Gupta\*, Eric Johnson\*, **Justin Payan**, Aditya Roy, Ari Kobren, Swetasudha Panda, Michael Wick, and Jean-Baptiste Tristan. Online Post-Processing in Rankings for Fair Utility Maximization. *WSDM 2021*. [link]

## Preprints

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\* denotes equal contribution.

- [P3] **Justin Payan**\*, Bálint Gyevnár\*, Atoosa Kasirzadeh, and Nihar B. Shah. greCAPTCHA: Assessing Understanding as Evidence of Research Authorship Under Generative AI. *Under Submission*. [link]
- [P2] Vishisht Rao, **Justin Payan**, Andrew McCallum, and Nihar B. Shah. ML Researchers Support Openness in Peer Review But Are Concerned About Resubmission Bias. *Under Submission*. [link]
- [P1] Sarina Xi, Vishisht Rao, **Justin Payan**, and Nihar B. Shah. FLAWS: A Benchmark for Error Identification and Localization in Scientific Papers. *Under Submission*. [link]

## Workshop Publications

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- [W1] **Justin Payan**, Nicholas Monath, and Andrew McCallum. Document Representations using Fine-Grained Topics. *Sets & Partitions Workshop at NeurIPS 2019*. [link]

## Code Contributions and Deployment

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- 2026 **greCAPTCHA** ([greaptcha.com](https://greaptcha.com))  
Public-facing platform for assessing understanding of research papers. Generates manuscript-specific questions, evaluates responses, and provides reports for human review.
- 2026 **PULSAR**  
Designed a telescope scheduling algorithm that improves astronomical objective completion while improving satisfaction of global scheduling constraints. Currently being evaluated for use at the ALMA Telescope.
- 2022 **OpenReview: FairSequence**  
Created and deployed the FairSequence reviewer matching algorithm on the OpenReview platform, used by over 50 peer review venues.

## Teaching

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### Instructor of Record

- Fall 2020, **CMPSCI 590N/CICS 580 Intro to Numerical Computing with Python**  
Fall 2021 University of Massachusetts Amherst

### Graduate Teaching Assistant

- Spring 2021 **CMPSCI 611 Advanced Algorithms**  
University of Massachusetts Amherst
- Fall 2020 **CMPSCI 121 Intro to Problem Solving with Computers**  
University of Massachusetts Amherst
- Fall 2018 **CMPSCI 383 Intro to AI**  
University of Massachusetts Amherst

## Mentoring

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### Undergraduate Students

- Summer '21 **Rudra Gopal**  
University of Massachusetts. Now: MS at Georgia Tech.
- Summer '21 **Gautam Chatterjee**  
University of Massachusetts. Now: Software Engineer at Wayfair.
- Summer '21 **Ed Almusalamy**  
University of Massachusetts. Now: MS at University of Massachusetts.
- 2023-2024 **Hassan Gudal**  
University of Massachusetts. Graduated 2024.

2025-2026 **Noemi Barbagli**  
Carnegie Mellon University.

2026-present **Yiming Shao**  
Carnegie Mellon University.

## **Master's Students**

2020-2021 **Eric Johnson**  
University of Massachusetts. Graduated 2021.

2020-2021 **Ananya Gupta**  
University of Massachusetts. Now: MLE at Diffbot.

2020-2021 **Aditya Kumar Roy**  
University of Massachusetts. Graduated 2021.

2022-2023 **Vedant Bhatia**  
University of Massachusetts. Now: Senior Software Engineer at ElisaAI.

2024-2026 **Vishisht Rao**  
Carnegie Mellon University. Now: PhD student at Carnegie Mellon University.

2024-2025 **Sarina Xi**  
Carnegie Mellon University. Now: PhD student at University of Wisconsin-Madison.

2024-2025 **Arshika Lalan**  
Carnegie Mellon University. Now: Senior ML Scientist at JPMC.

2026-present **Kerissa Duliga**  
Carnegie Mellon University.

## **Service**

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### **Research Community**

2023 **Workflow Chair**  
IJCAI 2023

2023 – 2025 **Workshop Co-organizer**  
Computational Fair Division, IJCAI 2023–2025

2025 **Workshop Co-organizer**  
Practical Improvements to Peer Review, AAMAS 2025  
Workshop on Incentives in Academia, EC 2025

2023, 2025 **Program Committee / Reviewer**  
ACL 2023; EC 2025; KDD 2025

2022, 2024 **Subreviewer**  
NeurIPS 2024; EC 2024; AAMAS 2022; GAIW at AAMAS 2022

### **Media Coverage**

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09/2026 **Times Higher Education.** Frances Jones.  
Researchers submitting AI-assisted papers they ‘don’t understand’.  
September 17, 2026. Coverage of greCAPTCHA.

## Invited Talks

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- 10/29/2026 **University of Pittsburgh, NLP Seminar**  
Hosted by Xiang Lorraine Li  
*Upcoming*
- 10/22/2026 **Rutgers University**  
Hosted by Arpita Biswas  
*Upcoming*
- 10/13/2026 **University of Washington, Data Science and AI Seminar**  
*Upcoming*
- 03/2023 **Brown University, E-GLAMOR Group**  
Hosted by Amy Greenwald  
*Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities*
- 01/2023 **Harvard University, EconCS Seminar**  
*Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities*
- 11/2022 **Carnegie Mellon University, Nihar Shah's Group**  
*Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities*
- 04/2021 **UMass Data Analytics and Computational Social Science Brownbag Series**  
*Envy-Freeness in Paper Reviewer Assignment*
- 04/2021 **UMass Amherst Theory Seminar**  
*Fair Reviewer Assignment*